MEDICAL AI · SYNTHETIC DATA

Synthetic data could solve medical imaging's data problem.

But are we measuring it the **right way**?

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THE PROBLEM

Medical imaging AI is short of data.

Scarce

Annotated scans take expert time, and rare conditions are under-represented.

Sensitive

Every image belongs to a patient.

Tightly governed

Privacy rules and ethics limit how data can be shared and reused.

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THE PROMISE

Synthetic data: images made, not collected.

Augmented

rotations, scaling, intensity changes

Simulated

physics-based models of how scanners form images

Generated

GANs and diffusion models trained on real data

Hybrid

combinations of the above

The hope: ease **data scarcity, class imbalance and privacy** concerns.

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THE FIELD

I mapped the field into six areas.

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THE CATCH

Most studies judge synthetic data by the scores it produces.

Accuracy, DICE, AUC: did the model improve?

But a better score does **not** show the data is clinically valid.

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WHAT THE SCORES MISS

Four blind spots

Clinical realism

Looking realistic is not the same as being diagnostically realistic.

Generalisation

Gains may not hold at new hospitals or on new scanners.

Bias

Biases in the source data can be replicated, or even amplified.

Privacy

Synthetic data can still leak patient information.

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TWO STUDIES, ONE GAP

Two different bets on synthetic data

Gao et al. (2023)

Simulated X-rays from CT scans, mainly to **replace** scarce real data.

Al Khalil et al. (2023)

GAN-generated cardiac MRI, mainly to **complement** real data.

The shared gap: both are judged by performance metrics, with **no expert evaluation of the synthetic images**.

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WHAT'S NEEDED

Evaluation beyond accuracy

Standardised evaluation

Shared frameworks, so results can be compared.

External validation

Testing on data from other sites, scanners and populations.

Clinicians in the loop

Expert judgement of clinical plausibility.

Clear regulation and consent

Rules for how synthetic data is made and shared.

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THE TAKEAWAY

A better score is not clinical progress.

Full article and PDF:

github.com/dgelbier23/medical-ai-writing

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